

ActInf GuestStream #037.1 ~ Baba Brinkman

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hello and welcome it's February 28 2023 it's active guest stream number 37.1 and we're here with Baba Brinkman this should be a great discussion an emergence Baba thank you so much for joining let's just jump right into it what brings you to this day how and who are you and welcome thanks thanks Daniel so much for having me here it's a real uh it's a real honor to join you and to uh you know dip into this community which I've been aware of for a while and uh you know like to try to follow the work as much as I can but I'm not a scientist myself uh I'm a rapper I have been uh exploring academic topics in hip-hop for quite a while I did my Master's thesis on Chaucer's Canterbury Tales I was a Renaissance and medieval scholar uh and I I rewrote the Canterbury Tales into rap and I used to tour around the UK wrapping My Wife of Bath's tail and Miller's tale Renditions which was a lot of fun and then at one point it was around 2007 or 2008 uh an evolutionary biologist a geneticist saw my Canterbury Tales show and he reached out to me and said I love what you're doing for medieval literature popularizing and popularizing it with rap but there's other topics that need rap popularization I think his exact words were would you be willing to do for Darwin what you did for Chaucer and this was a in the run-up to the uh the Darwin Bicentennial in 2009 it was the 200th anniversary of uh the birth of Charles Darwin also the 150th anniversary of the publication of the Origin of Species I was of course a big fan of Darwin and of evolutionary theory but not very expert so I said to this scientist yes I'll write some raps about Evolution but I could use some guidance uh he sent me a whole bunch of suggested reading sources books uh peer-reviewed papers everything so I took a sort of six month Deep dive on the state of evolutionary theory uh absorbed a whole bunch of books and uh and wrote These Rhymes but his uh his criteria was your performances will be at academic seminars with experts in darwinian evolution across all kinds of social domains and psychology and everything so you need to make sure you represent the theory accurately which means you have to run your lyrics by me so I can check them for scientific veracity uh which I thought was a fair criteria for that kind of context so I sent him my lyrics he gave me some uh you know some revision suggestions and then when I got up when I was like set to perform the premiere of my Darwin

show in February uh on Darwin day February 12 2009 uh the scientist his name is Mark Palin he was a bacterial geneticist he had this great intro typical British tongue-in-cheek style he was like I have fact check the lyrics and you're about to witness the first rap show in history that's peer reviewed now like peer-reviewed rap that's a great idea uh totally totally marks coinage uh but then right away the wheel started spinning like what other topics could you apply the peer-reviewed uh wrap lens to so I did my Evolution rap show for several years it was called the rap guide to Evolution it played Off Broadway won an award at the Edinburgh Fringe Festival and then other scientists started tugging me on the sleeve like oh the Darwin wraps are cool but could you do um Wilderness conservation and threatened species ecology could you do climate change could you do Neuroscience uh and and various uh great collaborations came from that so I now have a whole series of rap albums rap guides to various topics and my encounter with the active inference uh community and research base was when I was working on one called the rap guide to Consciousness and uh Professor Anil Seth is one of the uh I think main proponent opponents and contributors to the Bayesian brain active inference research Paradigm today and he was one of the main science advisors on the project so he was giving me Bayesian stuff to read including Andy Clark's what's next highly technical uh review of all the literature which I slogged my way through as a non-expert and ended up working Bayesian ideas quite in a detailed way into the rap guide to Consciousness show and what really resonated for me honestly I had never really paid attention to Bayes theorem or or Bayesian reasoning at all uh but I you know it took me a while to get my head around it but then I was like okay this is actually just a great uh paraphrase of the scientific method and I was already a massive proponent of the scientific method and I was geeking out on the degree to which the brain may be a Bayesian engine at the microscopic level producing a rational inquiry process that replicates the exact same Bayesian algorithm at the level of interactions with Society so I did one song in the show called uh good Bayesian let me show you how to be a good basy and change your predictions after taking information in and if you're thinking I'll be less than amazing let's adjust those expectations and I was right away thinking about like how how is a rap performance itself an instantiation of a Bayesian algorithm with the audience's expectations being revised in real time based on the rapper's performance and the rapper's expectations when creating the script of the performance about what reaction they'll get from the audience which is a prior belief about the effectiveness of the performance but when it's only a script don't actually know what's going to work and then you do the performance you see what gets the reaction you revise your priors and you also

revise your script and that was a concept that I had already been rapping about in my Evolution show I have a show a whole song and that called performance feedback revision which instantiates the kind of evolutionary natural selection algorithm but at the level of creative writing and performance you write the script you do the show you see how people react you go back you change the script and that's kind of like the genome being revised under phenotypic variants and how they end up Surviving or not surviving reproducing or not reproducing so um you know this is I I just started out rapping and trying to master the craft and each scientific topic I get into I start going oh this actually helps me understand what's going on in rap much better and especially actually one of the examples I play with in the show which is a lot of fun is like uh call-in response expectations of the audience like a perfect kind of demonstration creation of a Bayesian moment when you go you know Double Take look twice calculate the odds roll the and when I do that the whole crowd would be like dice you know but the expectation of what the rhyme couplet's gonna be is a game rappers play all the time like finish the finish the bar right and that is entirely around a setup of an expectation and the Fulfillment of it but then sometimes you can play with it and be like yo perceptual Basics ask Morpheus we live in the and I usually get the crowd being like Matrix back of me but then other times I'd make it more difficult like yo perceptual basics ladybugs like to eat and then like maybe one person out of 100 would be like aphids and I'd be like yeah you know that that was the person that had the correct scientific Bayesian prior is to conclude my Matrix Basics aphids rhyme pattern but usually only a few get it and then everyone else is like ah there's the nerd over there or whatever so you know it's it's become like actually informed my creative process in terms of how I try to set up and violate expectations in real time psychologically with an audience when I'm performing um and I will be honest that my close reading of the research was several years ago at this point uh the rap guide to Consciousness came out in 2017 uh 18 I was performing it Off Broadway 19 I was uh touring it around uh theaters in the UK and still performing Off-Broadway and then 20 of course all performances ceased all together and I had to think about what rappers do in pandemics when they can't get on stages and see crowds so uh yeah we can talk uh during this interview about my new project which is a custom wrap agency you can see me wrapping it right here event wrap uh which could be fun to talk about in terms of the event inference uh elements of active inference and and Bayesian analysis could be uh you know I think some of the aspects of what attract me to events What attracted me to events during the pandemics and still does as a sort of a catalyst for good art and for rap might have to do with active inference uh expectations

and experiences but I don't know this literature very closely at this point so um yeah that's basically brings us up to the present hopefully that was a good summary amazing uh just great to connect and to be able to share this moment with someone who's seen it from all sides in front of the stage on stage behind stage Inside the Mind real recognize real so there's a lot out there and just like as if we were talking about a paper namaste real recognize real namaste if we were talking about a paper we would read between the lines and reference out and similarly there's so much that can be referenced out and people can listen to the original creations and see what that primary text looks like and the fact that it's uh orally performed text for much of human development in history was our real so whether something is written is increasingly fluid these days with speech to text and text-to-speech so there's going to be so many great Avenues let's just start with event-based cognition so what is an event from the cognitive perspective I mean I was trying to find that definition with Google about 20 minutes ago so I didn't sound like an idiot in this podcast so uh I think it's a uh you know a it's a perceptual construct right it is the perception of an entity that there is some kind of instantiation of action that is circumscribed in time that has a beginning and an end and if it's not perceived as having a beginning and an end by some cognitive entity historically organisms but in the future also probably AIS uh then it's not you know then it's not treated as an event which means an event could be a thousand years or five seconds or whatever it's actually interesting how the kind of the scientific definition of it in active inference research really is like a narrative definition in a way right it's like it's like it's a story something you can tell a story about something you can construct a story around uh is an event is that pretty close to how scientists think about it yeah I mean already so much there you're right it does not have to be tied to a given amount of clock time which we might call chronology hearkening back to Kronos that sort of decimal point time but action takes place in the understanding of the actor or the observer in this time scale of action or Kairos and that time scale of action may be narrative and that is what you invoke with the beginning middle and end rising and falling cultural patterns that's not necessarily happening on the clock just because you can run a stopwatch while somebody's reciting Canterbury Tales doesn't mean that it'd be twice as exciting if they spoke twice as fast or that you could map in some linear way what was happening on the watch face to what's happening in the mind of audience who have different familiarity with the material or a historical study context as well it just made me think of like if I do a 10 minute long wrap of The Wife of Bath's Tale then that's an event I performed the wife of bathdale there was an audience that witnessed it you could circumscribe it that way but

you could also you know so you could do a scientific study on what occurred during that event right but you could also choose to study the event of humanities retelling of the story of The Wife of Bath's tale which is a story that takes place in King Arthur's Court so now we know where the mythology has to have started which is around you know forgive me medievalists but let's say like four or 500 A.D and then whenever the historical King Arthur's Court existed and then there was an oral mythology that adhered to that over the course of a period of time as people told folk tales about it and then in the year 1400 uh Jeffrey Chaucer uh well I think he perished in 1400 so he was born in the 1330s but he writes the wife of bath tale at some point as a written representation of the oral narrative and then that Vellum manuscript is passed down into college reading lists and I read it in 1998 and write a rap version of it so any study of Humanity's engagement with the story of the wife of Bastille has to really start uh you know in the year 400 and and the last time I wrapped my rap version of the wife of Bastille at the Edinburgh Fringe Festival which was a couple years ago and no event that accounts for the phenomenon of the telling of that story can leave any of those instantiations out right awesome it's like the earlier events are our prior or our generative process which is our Niche that events happen in our generative model our cognitive model exists within the niche the generative process and those prior events quite literally are our priors and there's no last chapter because the observations keep rolling in and whether the observation is exactly as expected whether that performance of that symphony is exactly as expected but they're that performance of the hamburger or of the portrait even if it's exactly as expected it still is a new observation and then what is as expected well that's in the encultured cognitive mind and that is this sometimes implicit reference point that novelty is contrasted against like well I knew I was going to get a rap summary but I didn't know it'd be that funny or I didn't know I was going to get a rap summary these are like different levels and types of novelty about the story of a moment or or look at how he changed the story from the archival version because there's definitely a completely different experience of people that are intimately familiar with trousers version and ones who are hearing the story for the first time and encountering each of the story the characters this is the kind of expectation of how accurately will be the story will the story be recounted in this new version uh also let me just um expand on something that you said he said there's no last prior because there's constant reinstantiations and updating right but there is a last prior from the perspective of a temporary analysis of an event you can sort of you can set a sort of artificial beginning and end for the purpose of doing an analysis and that's I guess where you're able to apply mathematics to it but you always have to like you know keep in

mind that it's a chosen beginning and endpoint within a larger unfolding yeah and that's very um resonant with my experience as a researcher which is like there is a final data point in any time series at some point the fmri machine or the ant foraging watching machine you just turn it off for the day so there is empirically something that's very addressable but in the general case it's not the case that a Time series algorithm says what the last observation is and we might even say that that's a deflationary perspective on history because rather than debating whether somebody has released the definitive American novel or the last word on this Symphony it's like well the last note of their Symphony was non-controversially the last another Symphony but the question about the discourse in the community of course there's no last word in this open-ended space and art is always like surfing on that open space one of the things I remember from my uh comparative literature courses is this idea of the anxiety of influence and how poets have their uh I think it's maybe Harold Bloom that wrote that but it you know poet poets have a sense of needing to create something new I think it's General like applies to Art all artists right theater rappers you know you know this is like imagination conjures but of course great art is produced on the shoulders of a lot of influence in the past and one of the measures in an English literature course of whether a poem Rises to the level of worthiness to be studied at University is the degree of impact it had afterwards and that can sometimes be measured by how many imitators it generated so one thing I always find fascinating is you read 14th century literature and Chaucer stands out as the main poet right if you barely read any 15th century literature in an English Lit course or at least when I was in school in the 90s that was the case I've just gone all blurry haven't I I'm gonna see what's going on with my camera here um there we go refocused the camera has its own priors um so uh you know like Chaucer shut down English literature for like a hundred years and when I was when I was in an English lyric survey course I'm like why do we read Chaucer in the 14th century and the next Port we read is the 16th century what happened to the 15th century and my professors were like well every poet that wrote poetry in the 15th century was seen as so much of a Chaucer knockoff and derivative because his style was so influential that none of them rise to the level of having contributed anything distinctive until over 100 years later I was like wow now that is impact right that is like as much impact as you but ironically it's like negative impact like the most significant poem shuts everybody up because they don't dare to do anything except for imitated in a totally derivative way but creating something new out of it you know that's what is creates the most fecundity in art right like you create something and a lot of her people were like oh I love this I'm going to take it and adapt it run with it that's one

of the things I love about hip-hop is it's so much about sampling and flipping and remixing and it's not afraid of any other genre because its attitude is you can take any genre and make it better by adding beats and Rhymes uh but but really effective art shuts it down I think Andrew Marvel was also known as like the last Carpe Diem poem writer because he wrote Such a perfect Carpe Diem poem Gather ye rosebuds while you may that no one really bothered with that genre anymore because nobody could do as well as Andrew Marvel and he wrote the last great Garden poem as well English garden poem and it's like you know how do you measure your success as an artist by the degree to which you've created something that's so intimidating for everybody else that they don't even bother with the genre anymore and now you're measuring your success by the deletion of genres from the cultural milieu which creates space for other people to invent more genres but this is an interesting way to measure success wow I'm gonna pick up on that pattern and bring it into attention and the Bayesian perspective on attention which is sometimes equated with precision so if we're paying a lot of attention to a sensory input a lot of attention High Precision we're really putting a lot of credence in that information it's overriding our priors a lot we're updating our beliefs to basically be what we're paying attention to and then the other case would be we're not paying attention we have low Precision or high uncertainty and we're not updating our beliefs at all that's what we basically conversationally mean when we say we're not paying attention to that like I couldn't tell you what it was it didn't leave an impact and when you were talking about this cultural evolutionary process I thought about cultures paying attention to genres and works because if there was one work that was the object of attention it would be in the spotlight but attention is like a spotlight and if it's a laser on one thing it's not on everything else and if there's no work that's rising to then there's this proliferation of hypotheses cultural endeavors but no one of them will necessarily be shared and so art that resonates with communities and with your friends it's like in this gray Zone where it has to have enough attention on it so that there can be a culture of hip-hop or the Grateful Dead or whatever it may be but not one that is so attended to that it gets burnt by the laser yeah it's like if everyone's only paying attention to one thing then that's actually very stifling for cultural flourishing because it's you know you don't you don't get all the variance you get these like totally dominant moments you can see that at the macro level in history but it also just happens in social media all the time too right everybody's talking about one thing and then it's like you know what what if that thing is not interesting to you or you know what that what is the upshot of that com of that dominance of attention in active inference we sometimes talk about regimes of attention and there have been

some great recent papers on cultural regimes of attention and that kind of takes that basal Bayesian idea of attention as Precision to stimuli and it puts it into the dynamical space where the regime of attention might be like the sequence of eye fixations across a page so that is a complex pattern with attention moving and we could just call that broadly like a regime of attention or attention and action and in some sense that's what culture is or can be described as as regime just the kind of summary of where you happen to look or is the intent that it's like there is some kind of intentional control of attention that is inherent to the work that's being attended to like you can expect people to look others in the eyes when you're seeing a video or a photo in the regime is that like there must be some implied like uh maybe in inbuilt psychology of things that we tend to find more fascinating is that is that what's implied by it or a whole lot there the word regime does have a slight connotation of something that's imposed whereas it is in the neutral sense describing merely the Dynamics and then what those shared regimes of attention might be like we're going to approach each other this way making this kind of eye contact and make these kinds of gestures that's how we're going to be reducing our uncertainty about who we are and what this event is and without those shibboleths secret words and implicit or explicit processes and rituals it's hard to lock on culturally there's so many ways to go from there but I'll leave it yeah I'm always whenever I see shibboleths and Norms being acted out I'm always trying to I mean obviously there's scientific studies that are trying to unpack this as well but my favorite thing to speculate about is to what degree is this culturally constructed in an open-ended way and to what degree is the cultural aspects of it uh relying on an underlying cognitive psychology that is inherited From Evolution like for instance you know if you're having a conversation with somebody one-on-one and you never look them in the eyes you know you that's inherently suspicious right like what's wrong with this person they won't look me in the eyes are they lying to me do they have something to hide like that response that you get emotionally and in terms of like somebody's disposition towards you you I would find that very unlikely to be 100 culturally constructed like could you imagine a society that was egalitarian where they never looked each other in the eyes I can imagine societies where Aristocrats prohibit uh people below them in the hierarchy from looking in the eyes because it's seen as a sign of disrespect which of course is entirely based on primate behavior when you're like challenging an alpha if you look at them for too long then they'll bite you and you gotta like cast your eyes down to demonstrate your submissiveness or whatever like humans have replicated some of these very deep mammalian instincts that we have right but like are we malleable enough that a society could ever evolve

with the norm of nobody ever looks the other person in the eyes or would that street swim upstream so strongly against our instinctive ways of measuring our degrees of trust and Affinity towards each other that that culture would just inherently fail and and not manage to be represented in the panoply of cultures going forward I'll try to give a thought on that falling back to this attention discussion so the movement of the eyes does provide information whether it's deceptive or not it is an outlet of information from the person who you could be looking at so it's putting attention itself right where you look to find out if someone's paying the eyes are the window to the soul maybe that wasn't just about the cornea attention as the window to the soul it was about Game Theory and so to neglect this Fountain of information you have to have a cultural clamp on it like a secondary reason to not look because it is providing so much info on a millisecond by millisecond level in ways that aren't always even expressible like pupil diameter has a lot to do with arousal and psychologically people may or may not know about that empirically and I doubt it was shown empirically until we had cameras and measuring devices and millimeters and all of that but it can still play a role implicitly in behaviors and heuristics and strategies and so it's like if we're paying attention to our embodiment certain things will arise in situations where the inherited prior has to overrule or reinterpret those first principles takes those are like scenarios with an attention to the Past for better and worse that is shading our experience interpretation versus the Prima fascia situational awareness that convergently evolves to a few attractors like looking at informative sources of information yeah and and just because there's a lot of uh swimming Upstream of uh sort of instinctive cognition necessary for uh cultural norms to play out doesn't necessarily mean that they're doomed to fail or you know a waste of time or whatever you could almost say like the whole scientific Enterprise complex mathematics even reading none of these things are you know instinctively evolved they're all very recent and we're like building them on top of our uh older cognition and and architecture and instincts but they're you know gloriously culturally constructed and I would say like worthy of uh the effort of swimming Upstream the keyboard q-w-e-r-t-y oh well the keyboard doesn't have to be that way well yeah but we don't need keyboards and we haven't always had them and we won't always have them and left to right reading it could be right to left or up to down or it could be any but we haven't even had inscription technology so maybe let me bring it back to your experience with hip-hop how has the generative process the extended cognitive niche of Technology supported the remix culture and how does that manifest well I think I'll start with some of the good ones I used to go to a studio and sit with producers and hear their beats and uh you know like let's go

with that one and then like Try lyric side pre-written or sometimes write the track um it was really expensive fun interactively I I do miss the in-person studio experience but all the previous rap records I've made for the past 10 years were all made remotely with Dropbox so the rappers you know send or the producers send me a folder full of Beats I choose the one that I like the best I write the rap I record the wraps at home I send them to the producer and I've got like a track where the producers based in Seattle the uh session musician is playing Keys is in Vancouver the vocalist is doing the song uh you know from his place in Toronto and I'm in Long Island like taking all the different parts and editing them and then sending them to the mix engineer and the master you know it's like this great distributed collaborative process and you know that's even by today's standard that's analog right that's like sharing files in a Dropbox and throwing them into your local studio and and sort of like measuring each input against the expectation a song to sound ultimately and giving people feedback and asking for revisions uh but I think there are now ai uh sort of web 3 platforms where you can all be in the same virtual room and the latencies worked out in your collaborating musically I haven't used those yet but I've been uh you know I've been scoping them out so that's been a very I think that's generally been positive it's brought the cost down and it's allowed for real time well not quite real time but like ongoing collaboration with creatives across different areas uh but then the kind of parallel Revolution that's happened is from selling CDs uh selling tape selling CDs and then Spotify takes over and you're getting like fractions of pennies on the stream and your music is going into this massive Archive of every kind of music and you're only getting paid based on how much your music is listened to compared to every other music that's on there they're partitioning their subscription fees and it's really it's become much harder uh I think for a lot of musicians to make a living from the sale of music and so over the past 10 to 15 years that process really pushed a lot of artists into really just basing all their income on live performance like setup gigs and and Rock shows and sell Merchant shows if you want any expectation of income from music sales you better be there meeting the fan shaking their hand and sending them a CD in person then kovid shut that down you know for a couple of years anyway it's coming back but my estimation during the pandemic was even as it comes back it's gonna you know there's gonna be people that are more reluctant to travel to shows and you know do that have the in-person experience even when Public Health notices say it's safe because people have different degrees of paranoia and anxiety and they won't feel as comfortable even if it is objectively safe uh so you know pivoting to this uh event rap game was partially based on an analysis of what's happening with the industry and how hard it is to

make it as an artist and part of that has been uh you know a technology driven process wow also many ways to go let's kind of return to those expectations and I think there'll be some ways that your experience with my map or resonate to the Bayesian brain so say a little bit about what are your expectations when you're coming out to a performance whatever setting that you choose to think about and then how does that interface with or to what extent do you even preempt or pre-game the audience's expectation do you just call it like you see it or are you looking to finesse your performance based upon what you rightly or wrongly believe their expectations to be well there's an ambient uh characteristic that I kind of just take for granted which is that I'm not very famous but I have you know a core hardcore Boutique fan base of the like science rap science nerds and science rap nerds that so if I'm in front of a crowd most of the time more than half of them don't even know who I am and I've got to introduce myself to them for the first time and some portion of them have seen a bunch of my shows and their fans and they're like oh it's his next show you know but I don't really write to uh address an audience that knows me generally and my assumption is that they don't and I'm constantly reintroducing myself and finding new fans that way and let's be honest like you see me walking down the street you're not going to be like there goes a rapper you know the Bayesian priors of my ability to perform as a rapper are low and they have been since I started I was as bookish you know Suburban Canadian woodsy teenager like I want to be a rapper but no one's gonna believe I'm a rapper so I better get really good at rapping and also you know just it is a bit of a choice not to like dress overtly to hip-hop or talk or overtly to hip-hop it's like there is a hip-hop Persona that a lot of artists adopt maybe becomes natural to them and they could say this is who I am there was either a conscious or unconscious decision at some point in their growth where they you know became more seemingly hip-hop maybe to reduce the cognitive dissonance of the lack of uh you know expectations of them being able to rap from Their audience maybe for other reasons maybe peer influence maybe they just you know none of that analysis really uh matters and they just they felt it they were passionate so they started adopting the lingo and you know I I kind of uh always partitioned rap a little bit like when I'm performing I'm a rapper and my profession is I'm a rapper but I'm not gonna like make myself a cup of tea in a rap style you know what I'm saying is like I I do distinguish a little bit between the domains of my life and uh what that partially what that means is people don't Mark me and expect me to be good at it uh or to be able to deliver and then I get to you know there's a cost benefit there right for one thing like the skepticism might lose me potential opportunity entities to perform or potential members of the audience who like see me setting

up the mic and they're like that guy I don't think so and then leave but on the other hand the transformation I get to take an audience through as they realize what they're experiencing and and feel their uh expectations being overturned is actually I think really part of the joy of the experience you know it's an expectation shattering experience when someone that you don't expect to be able to deliver in a certain art form just really does and uh and I and I kind of play with that in the comedy and tease the audience for having low expectations of me and stuff like that um you know then but then uh part of this is contingent on me just never getting that famous you know if there was a point at which I had achieved as much as much of a fan base as mm my like writing style about you don't think I'm going to be able to rap do you while watch this would of course sound pointless kind of like he stopped writing about being poor after his first album and the first line of his second album was they say I can't rap about being broke no more you know just like naming the critical uh reception of his transition from album one to millionaire album two uh so you know I think there is a kind of interplay between artists and their audiences and you know I I I stay Niche and I write to the expectations of that niche we're all in our own Niche so I hope that part doesn't change but very interesting when we attend we don't even need to say pay attention and use this economic modality but when we attend to our experience and update our priors just statistically that's called learning and I think that speaks to this synthesis with the language the Lexicon the slang or as we might call it the active inference ontology speaks across a Continuum from the experiential and the folks psychological the day-to-day it's the same terms with the same authentic relationships that are increasingly becoming embodied in our intelligence algorithms again not even calling it artificial intelligence algorithms because their extensions into the niche from us so these cognitive models that are being deployed AI artificial intelligence or large language models they are cognitive entities and they're participating in these ecosystems of engagements with us so I uh I had a conversation with the new um chat GPT driven Bing engine last night I just got I just got added to Bing uh the the AI driven version it's interesting the first thing I did when chat GPT came out was I said write me a rap song in the style of Snoop that explains how to change the oil in a Honda CRV because that's exactly the kind of project that we do with event wrap you know event wrap is like a team of skilled professional 10-year veteran artists that'll make you a song on any topic and either record it as a video that you can share uh as a kind of public engagement or explainer or come in live to one of your events virtually or in person and perform but it's like top it's a topical rap agency it's like all the rappers have their own catalogs but they don't perform them at the events we book they

always create all new stuff and uh chat gbt's Honda CRV wrap oil change wrapping the style of Snoop was like uncannily on point it was very like it you know it had the like Snoop Dogg lingo it was like an accurate description of the steps that you would have to take uh to change the oil in your CRV it was like a cogently you know from a scripting perspective I felt like we had a threat on our hands you know like Chad gbt has definitely like shook a lot of people's uh professions marketing copy professors academics it's writing papers like yeah it's you know the economics of this thing have not even begun to Dawn on people yet but the question in my mind was how soon is it coming for the rapper's job and uh there was there was a sophomoric cartoonish kind of kindergarten flavor to the way it's constructing bars uh you know so chat gbt and and Bing uh by extension I think it's the same uh uh layer that's running it is um you know it's it writes Raps like someone who's never wrapped before so if I told you write me a rap you could do it but there's like layers of proficiency of you know Master Level stuff that that of like syncopation and internal Rhythm within the main stresses of the bars and especially and this is where I was chatting about with Bing last night is polysyllabic rhyme so polysyllabic rhyme schemes are like the construction of multi-syllable Rhymes out of parts of other words and I was like um uh asking Bing can you do polysyllabic Rhymes and Bing was like yes I can and I said okay give me some examples from mainstream rap of polysyllabic Rhymes and it pulled an example from MF Doom which was is an excellent example it was like um you know the girls are crazy schizo but she's got amazing tits though uh so you know you can hear the pattern crazy skitso amazing tits though it's like multiple words constructing a longer pattern of syllables I was like gravo bing you found a perfect polysyllabic Rhyme now write me a wrap that explains your internal uh data architecture or your algorithmic architecture in the style of MF Doom emphasizing polysyllabic Rhymes couldn't do it all monosyllabic like check me out I'm being I do a verse I am artificial intelligence I don't even need to rehearse and I was like bing those are not polysyllabic Rhymes you started out quoting MF Doom complex five syllable rhyme pattern and now you're rhyming rehearse with verse like don't insult my intelligence and I was like dissing the AI and then I was like I'm sorry I'm a new model and I will get better at polysyllabic Ryan but basically that is the thing that's got our job safe for now because if rap is an elite art form that takes a decade or more to master the techniques of flowing live freestyling polysyllabic rhyme and Rhythm patterns and also there's a confidence to perform are the things that we're still getting paid for even if ability to write a basic like competent rap is now easily achievable by AIS I'm sure even outside of the rap profession these conversations are happening they certainly are in research and education

previously the artifacts that we exuded into our Niche were mnemonic almost at best it was probably transformative when some of these oral Traditions were able to be written down and that may have also been seen as threatening to the underlying basis of the skill and the transmission and now we had just a brief window of these inner externalized artifacts these mnemonic devices have only existed AKA writing for however long they've been in the niche and now it's like they've come not quite to life but certainly to activity and now these off-sourced and outsourced cognitive entities are not just receiving and holding but doing and that transforms who and what we are in terms of our perception cognition and action loops they're going to be different and that is what any given particular thing is so you mean like any any technological innovation that changes how we interact with each other and with our art forms yeah and and from the technical side we use the word thing in active inference to describe the blanket and the internal States so we take a particular partition we cleave off the thing we're talking about from the environments the generative model of the thing is the blanket and the internal States and so that thing is defined in terms of its perception cognition internal States and action so when new sensory modalities or sense data appear when new action modalities affordances appear and when our thinking is different it's a different thing it isn't just like well we mixed this one thing with another thing and they both retain their identity it's like there's been transformation and a learning and were in the middle of some fractal dimension of it yeah one of one of the uh I mean I think this is like a radical break from from previous technological changes you know one of the things I argued in my thesis um when I was a master's student was that poets conceived of Their audience and their ability to connect with people orally primarily in Chaucer's time right reciting your poetry for a listening crowd but then the invention of the printing press with Gutenberg changed that ability to imagine an audience because it was so much easier to imagine our silent readership of 10 000 people reading your pamphlet but making ten thousand Vellum manuscripts by hand was always prohibitively expensive so no poet could even imagine ever having a fan base through text right Gutenberg allows for the possibility of a fan base through text which changes the whole art form suddenly rhyme is extraneous because they're going to be reading quietly anyway so who cares what the Poetry sounds like the main thing should be carefully constructed syntax and story in semantics in a concentrated way but you see then this rise of free verse right Shorty's poetry is sort of like shorn of all of its ornaments and my hypothesis was that they were just basically irrelevant because people were reading quietly and not reciting out loud and then uh Edison invents recorded sound which creates the possibility of an audience and

fan base that's just as vast from the sound of your words as it would be from the silent readership and that paves the way for the possibility of hip-hop and because this is like people that are writing things rhyming verse and of course I call this the rhyme Renaissance right poetry was all rhymed then none of it was rhymed and then if you think of rap as poetry now it's all rhymed again because who cares about poetry rap dominates like all of our verse consumption compared to every other kind of poetry out there right and rap is rhymed and I think people instinctively resonate with rhyme and just took a break from it when they were reading silent pamphlets published by Poets who were imagining their readership but each of these examples is like a technological innovation that mediates the creative process with its ability to connect with an audience but this AI thing is something completely different isn't it it's like competing with us for it's a in our production of the creative things and right now I'm taking some comfort in the fact that it's competing ineptly it's not as good at writing a rap as a good rapper but you know I'm probably a subset of professions that can still say that and lots I mean it's passing the LSAT it's passing the um all the Masters entry exams and you know which which rarified skill sets will be left maybe I could comfort myself to say the rapper will be the last to fall to AI because of the combination of creativity and semantic cogency and humor and all the things that go into it that the AI is going to have to work each of those out like maybe stand-up comics too uh are safe for a while because GPT can write some jokes but they're corny jokes still right like original jokes it writes aren't that good compared to you know a Dave Chappelle so but but what are we talking like 10 years of safe jobs uh as artists or less you know like the speed of this last six months has been a little mind-blowing who is the deep blue who is the Gary Kasparov of performance what is the Turing test and for you to again calling back the real recognized real I listen to that as being like kind of a Turing test is you have cognitive entities in this moment of reflection which has been called theory of mind and in active inference sometimes is called thinking through other Minds ttom because there's this cognitive emulation of if I do this then they'll do that and so thinking through other Minds used to be a defined set of types of Minds we had children we had adults like us we had adults in different cultures we had squirrels and then the workshop is open and the composability is Extreme especially with algorithms that can construct other algorithms and all of a sudden that ecosystem of shared intelligence thinking through other Minds we don't even know if it's a sports car or a horse and buggy or a spaceship and someone's like do you want to get in and it's like where are we going what is it going to be like all those expectations that we might want to have about a journey and an event and who can say what with any Precision things

will be like so what do we do then yeah except it's more like a spaceship that people that the pilot is like do you want to get in if you don't all these people are gonna die uh and so you know it's it's your choice whether you want to get in but your lives on the line because obviously people that work in all kinds of medical fields and scientific fields will point directly to the amount of human suffering that they're on the verge of alleviating using these tools and the applications to wrap our job stealing are merely you know side effects spandrels of the main focus of it which is I think mostly very laudable and very necessary right like this like let's not get on the spaceship means you know we continue to put up with all kinds of genetic diseases and you know levels of poverty and public health crises and things like that that are being that the AI is being applied to and is likely to do a great job in solving yeah someone says look if your four-year-old still wants to do art by hand we don't need to stop that but we just want to use it here that's going to be something that's very challenging to take a long-term perspective on because it's going to seem like a reasonable compromise in the moment just like everything always does and what compromise was made when those oral traditions were written down what compromise was made when that sheet music became recorded and that became canonical and distributed and so what is the nature of the compromise and who's at the table in 2023 yeah that that I think that's exactly the right level of analysis and uh you can say on balance that each of the compromises we made for some new technology to you know give us some cost but also benefits it was a totally understandable and reasonable one uh you know from the perspective of like when I look at an individual poet giving up on rhyme because they want to do pamphlets and reach more people I'm like well they got a family to feed you know that's the way that they can achieve the notoriety and the claim that is their path to success as an individual and it has cultural ramifications overall and that you know that's I think you can put that like people can opt out of any technology and the people that decide to opt in have totally defensible personal choice reasons for why they're doing so usually to get a competitive advantage that was the case with you know tools going back to the Stone Age right I only throw rocks I don't use bow and arrows all right you do you I'm a take down this Antelope over here uh you know we'll give you a piece if you're lucky uh you know that's just that's that's the AI tool choice that everyone's facing right now too it also makes me think about art always playing this role of like the tip of the spear that kind of penetrates the Veil we had the discussion about nfts as if they were just a question about art but of course digital certificates are a lot broader and more significant than just visual images and there's a discussion about whether certain kinds of art are being replaced by generative

algorithms long ago six months ago 12 months ago when it was pre-chat GPT but there was a lot of the stable diffusion and midi Journey type algorithms then the conversation was more about graphical design and professionals but of course that is a broader discussion that isn't just about the generation of images that you are paying a creative professional to make and it's like art as this Frontier and mediator with the open-ended will people be looking for open-endedness and liminality in their events in 50 years I think they will be that niche the marginal niche of Art as well as the central role of Art I think it's fair to say we'll both exist now whether that cultural attention on the central artifacts will be about artifacts made by a person an algorithm or something in between you know if somebody designs a 3D sculpture but then it's printed or if they think about a drawing and then they use a pen that was produced by a machine who's to say in these extended scenarios whether one or the other made it but in the margins it's where we're always going to be exploring in whatever people call Art and people who are in the middle of the spotlight saying that's not art it's not their art every single genre has had to have this mitosis or birthing with people who weren't part of that process saying it wasn't art or that's not the right way to do this kind of science or rap is not music you know that as a hip-hop partisan I'm very familiar with uh defending against the rarefied tastes of people that want to gatekeep what constitutes art and what doesn't but um you know one thing I've always loved about rap that like that that aspect of it not being gatekeep by elitists you know like if you want to be a professional ballet dancer you know like Billy Elliot stories notwithstanding you got to try to get into Juilliard and have you have to go through the process of rarified Judges that are telling people who can and can't get into the next one Lake like there is no back door to success in some art forms right classical music Opera but then but with hip-hop there's no front door to success it's like you got to get good at it on the street level there's no degree that'll get you a job in it you just have to find crowds and entertain them uh at the level of your neighborhood and then your city and then your state and and if you're you know if you survive the levels of competition on the way up then you get to the you know the war World stage or whatever but it's very like you know pick up a mic and show what you got kind of thing and it's mediated by the audience and the fans uh you know if they like it it's right and if they don't like it is wrong and MC has moved the crowd and you know there's a fairly like definitive objective measurement about whether someone's doing rap well or not and is how the crowd is responding there's no valid argument that everybody's booing you but you're actually awesome because they just don't understand your art no you failed to understand the context in which your art was meant to be consumed right so whenever we're

looking at these trade-offs often it comes down to producers and consumers right each new tool is good for consumers but bad for producers and this is like the taxi guilds trying to fight off Uber Uber is great for people that want to ride but it's bad for people that are going to try to control the means of production and you're seeing this exact process play out with chat GPT now which is like you know and also with stable diffusion and the other examples you gave like people want to consume art and they don't necessarily care who made it as long as their reception of it gives them the feelings that they want from the consumption right and that I think that's the case with music and with poetry as well although we're not yet consuming a ton of AI generated music and poetry and comedy because it's not sophisticated enough to produce the highest level stuff although I think with classical music it is and with graphic design and art it is right like the best AI art is actually competitive commercially or you know from a consumer perspective you know like taste test wise with the best rarefied professionals which is devastating for them but it's not unique it's been you know that's what the guilds have wanted to do for centuries is protect their status as the elite producers of a thing from the ability of just about anybody to consume it for as cheaply as they want and that's what technology keeps delivering and in general I'm actually a fan of that process I think it's democratizing it's good for everybody on average even though it perpetually upends the guilds but we shouldn't give guilds control over our ability to consume excellence and if if Excellence is delivered better by AIS then let's not be curmudgeons about that and I can actually you know but but there is still an individual challenge like okay what am I going to do to make a living now you know it does really create a lot of potential unemployment and economic Devastation which is why a lot of people have thought that there's just no Escape From The Universal basic income because the jobs are going to fall and Society will descend into chaos with that much unemployment and frustration and resentment and anger but you know even if rappers are the last job to fall you know the analysis has to apply to all kinds of other jobs first and uh active inference comes into this because you know how does a rapper entertain the crowd by having a mental model of what the crowd's going to do if they perform action X right what it will be the effect of X on y why is the crowd X is the intervention and the intervention is a choice of a bar I'm going to rhyme this with this I'm going to make a song about this topic I'm going to try this joke out in my flow as a punchline and you have a mental model oh yeah people are going to think that's really funny you know and then you deploy it and you'd see the degree to which your expectation has been fulfilled and I don't think AI is that good at that yet because it doesn't model expectations of audiences very well yet does it it

doesn't have deep theories of Mind of like perceptions of perceptions of perceptions it just has averaging abilities to recreate patterns that it finds in vast troves of human created art which includes pixel patterns and graphics and and patterns and text romantic and syntactic patterns but like performance art with an audience requires a mental prior of the audience's disposition and attitude and that that's why I say rap's going to be the last to fall but and that's also why it's really hard to get good at you know it takes years of Performing dying on stage and sucking and being booed before you really build up this mental model of like I know what people like and I know if I do this they'll do that and even I've been doing this for 25 years I'm wrong all the time oh yeah that line fell flat all good all good you know either you like carry on or you go fix it that's where the AI That's where the AIS have to get to right yeah wow so much one thought was rap will be the last of fall in the first to rise around that campfire again it'll arise situationally you mean after Society descends into chaos and all the AIS are resting and we're back to Spears rap will be the first to arise again first in last out sorry go ahead what was the next thing another thought you mentioned taste makers and uh I thought what is the ultimate or even the proximate taste maker it's our generative model of taste it's every person eating that dish The Taste maker isn't who told them to go to the restaurant that's basically believing the hype and using the Paradigm of the advertising industry The Taste maker is the taster that's who is generating the experience of taste taste isn't just in the dish before it is actually eaten by that person it's experienced and they're going to have some previous taste in their mouth or they're going to have a context or a cultural understanding or a memory a trigger so the taste is generated authentically and situationally yeah there's going to be a lot of volatility as taste making becomes actually understood to be personalized and that that may look like the personal generation of media which has not even been plausible when you had only 33 RPM records then you have somebody selecting which ones to distribute but when you have algorithms it takes it one level further which ones you like to listen to people who like this also listen to that and this is like as far as we can see another level of like you liked this we're gonna regenerate that story so that the person's voice is like that that takes it to the next level but also with enormous trade-offs it brings it closer to how things really are well that that makes me think of something we spoke on earlier which is the degree to which cultural norms are able to reorganize or swim upstream against instincts right so when you say the tastemaker is the taster uh the counter examples that jump to mind are rappers who are like Jay-Z where he's like and before this song ends I'ma start another Trend you know like if you want to say the tasters of The Taste makers I'm looking at people who merely suggest something is cool

and it instantly becomes cool and everybody is able to post hoc rationalize reasons why it has intrinsic qualities that they now recognize as amazing over and above what the taste maker told them to do but we are highly susceptible primates uh to cool factors and you know so I would suggest The Taste maker is not the taster The Taste maker is the best predictor of the average response of the tasters and when you get very very good at understanding instinctively how people will react to The Taste using taste as a metaphor for all kinds of art then you're able to be treated as authoritative on taste because you're right so often about what people will like and that power of like vested expertise from your fan base then allows you to start trends that are not just predictive you say something's great and everybody instantly believes it's great and experiences it is great as authentically as if they had discovered it for themselves and decided it was great because they've recognized your actual real vertical average ability to predict taste across groups awesome I I thought okay again who's the taste maker it's the chef and not everyone's in the kitchen because a chef is in their position because they've been recognized to repeatedly be able to construct dishes that in the local taste making diners or maybe just a few diners like and it's like what you can't do that with Molecular Gastronomy but some so it's like it's the chef and it's the social influence Network and it's local what framework can we use to think about all of these interactions within and among cognitive entities within and among their niche we use active inference as just a way of thinking about it well what is active inference it's there in the title action and inference it's involving action and doing cognitive modeling and that gives us the composability and the ability to reconfigure and remix to capture all of these diverse scenarios not to close the book on them but to be like oh wow what an amazing question there's 50 phds of research that you could go into so it's a question generator and it's a lens by people who are even just beginning to learn it who review phenomena that they've already been told or experienced in one light all of a sudden they're seeing how it can be reconfigured for example seeing what we prefer in terms of what we expect rather than what we prefer in terms of what's rewarding which is the reward learning reinforcement Paradigm which proposes well maybe you know the value of different states and you're maximizing value it resonates with some contrast that with you have this complex generative model and you're always referencing to it and you're taking actions and learning to reduce your surprise bound your surprise of certain kinds which could include increasing your narrative surprise and you want that page Turner or the horror film so it just kind of reflects back and then one last point on octave inference as a research direction that the university you hear your thoughts on is it is happening in a Time that has never been seen before in research

and Academia so you said there's certain careers um or genres where there's no back door to success and then you said in rap there's no front door so traditionally even if we could say there was no backdoor to success in research the reality is there was a side door and people were getting kicked out and being brought into the party through the side door and a bunch of people were waiting at the front door they didn't even know that the show was not happening that way were developing a framework in community at the speed of pre-prints and live streams and so there's transdisciplinary collaborations with philosophy historians of science cognitive researchers mathematicians physicists so there's a transdisciplinary conversation and one with practitioners and one that's happening with everybody with view access to some large parts of it which wasn't even the case previously open science and research was like break down the paywalls everybody should have access to the PDFs which is great but it turns out that having access to 200 000 PDFs having access to every single PDF that Carl friston has ever been an author on it's not even going to take you 10 percent of the way unless you're gonna engage in this total monastic life to unpack them and the details aren't even in what's necessarily those papers it's beyond that it's in the community it's in the software in the inaction so of course wait and see but there's so much with the way that active inference is developing on the influence of previous Frameworks Bayesian brain and Bayesian statistics causal inference statistical physics happening in this sort of wrapping in front of the crowd way asynchronously with the pre-prints synchronously with remote presentations and recordings that it's hard to think of another area that has had that kind of a developmental niche you're saying active inference is the hip-hop of the scientific Community basically you said it not me live on the mic I love it yeah and it's uh you know I I think it's super exciting research and collaboration happening and it's I mean it's very very unknown to the public you know if you try to explain what this conversation is about uh you know people understand the idea of expectation and priors and belief upgrading the very sort of basic version of of the Bayesian algorithm but um you know all the applications of it and the cross-disciplinary side and everything you know it needs more science communication I've made two Bayesian songs I want I think I need to make 20 more because this is uh you know the the applications of the research are vast and especially you know in the AI era like people really need to know what these algorithms are achieving and the ways in which they are similar or different to our own cognitive processes and what we can expect them to achieve uh and how we can expect them to help and what they're threatening all that stuff goes through active inference doesn't it awesome well in our kind of closing segments would you like to do something musical or how

do you want to proceed yes indeed I'd love to I'm uh I'm very happy to do a demonstration of one of the last art forms that AI is not yet able to model and that is a freestyle rap so uh if we have anybody watching on the stream or Daniel you're welcome to do it as well if you want to pop some phrases into the uh the text I will look at the text on uh that you're displaying and also look at the uh the comments and the Chats on YouTube and anything that I see pop up I will visually uh process and convert into Rhymes in a freestyle rap track and uh you know just just to give a little insight into this this process you know it's like um it is a violation and uh and like uh expectation setting and expectation violating process right like the the active listener to a freestyle rap is trying to figure out the semantics and figure out where the wrap's gonna go and the wrapper is also in real time figuring out the semantics because it's completely unscripted but it's uh you know then and and at the same trying trying to set up expectations of the audience and violate them I think a lot of the fun is the surprise factor of like oh I didn't expect him to go there I thought he was going to go there but oh now I see and like it's kind of like a game of Chase like the rabbit and the fox uh so I see Sean Kelly's uh drop topology in the in the uh chat of YouTube I'm gonna start there and I'm gonna kick a little rap for you guys yo this is hip-hop tell me what you think this is is Bubba Brinkmann rocking now on active inference that's the inference looking for where info is I don't know but I feel it's kind of intimate yes indeed hip-hop what you think this is is Bubba Brinkman here on the active inference podcast I'm looking for where the info is so I could update my priors and get into it and maybe drop a beat Sean Kelly said topology yes right now I'm rocking hip hop ology can't you even see that's how I adjust the bass and treble topology that means checking all the levels and looking for the map and the different levels of impact I'm spitting rap yes I'm looking at the jet that's how I kick it yes how I go with rap and flows it is me kicking it with Mr Daniel Friedman house speaking yes I'm always just deleting variants every time I'm sometimes just weekend through the speakers I just do this on a Tuesday or a week and indeed active inferences how I'm speaking I'm sneaking past all your expectations like a rat and a maze because my raps will amaze and yes I'm doing Holly syllabic too amazed I'm just having to blaze yes I'm actively playing all the different rhymes on my brain like a piano changing the channels like I'm rocking gangster Tony Soprano yeah that's okay this is how I just anger the listeners sometimes cause raps are danger gotta give a shout out to Tucker Cahill Chambers who rock this with us yes Working The Angles like bass theorem can you even hear them when I'm Nero and I start to see it as this Mass scientific theorem that I could apply to every single single thing as Barbara Brinkman gets all your neurons jingling it's how I attacked it I

attracted a massive I'm active when I'm doing inference in my rap and indeed this is how I kick it when I enter my rap is directed it's never non-censored as I just try to constructed mistakes I make sometimes I'm Run DMC and sometimes I'm Drake except I'm not quite as good of a singer with The Melodies but then again I've got the freeze thieves that kick a lot of active inference Bayesian Styles freestyles that's trying to make the people smile less they're listening yes that's how I'm straight rocking even a mod mostafi says yo wow awesome with explosion emojis Baba Brinkman coming with the dope beats people are like hopefully he's one of those MC's that could Rock straight up as ancient hip-hop like Rakim and Eric B I never do it terribly I just free style that is improv which means it's spontaneous my style is the Zanies I'm repping for the long Islanders plus the Canadians people listen and blow their brains when I'm blazing it yeah it's how it goes when I'm speaking on the basics I'm in Long Island but Daniel he's up in Davis California I put it on you it's active inference Bubba Brickman when I'm speaking it I'm getting into it looking for where the info is that's how I'm rocking this was like a four minute continuous talk and demonstration yes indeed I'll make it easy let's see you do this [__] chat gpt3 you can't do it you can't freestyle over this music and use a brain to actively infer that's so fluid yes that's a wrap that I make up and invent the last four minutes has been an event prescribed in the time that's the freestyle rap that I did on this concept that's how I can adapt yes indeed that's a rhyme that I had to make I can do this my brain makes me a primate but I don't make mistakes okay sometimes I do but that's just mutation yes that's how I move all the mistakes get integrated my work is huge thank you Ahmad mustafi I appreciate you okay now I'm starting to wind down and trying to just find a new direction for the rhyme sound this is how I just get into it it's Baba Brinkman rocking with active inference Act give inference this is how I'm never pretending I'm trying to infer some way for the raps ending but wait a second this is how I do it hip-hop the easiest solution is just turn the music off foreign that's freestyle rap and I really appreciated this shot chat it was a ton of fun and super interesting I'll give a closing remark and then any last words from you it brought a smile and a laugh to my face and body because another individual was engaging in the shared moments I've seen language models output incredible X in the style of Y and every single time it brings a chuckle or less and it's like okay I kind of expected it to do that I want and um I'm disappointed if it can't fulfill my request because it's like well I expected I would I asked you this because I expected you to do that when there's joyful spontaneous engagements is reflective of a shared experience it's the real thing and when that lens is applied it's going to be real recognizing real and I don't think humans will be out of a

relationship role for each other for a long long time Amen to that and um I just want to say that the phrase that you just coined uh joyful spontaneous experience that's reflect or joyful spontaneous uh engagement that's reflective of a shared experience uh that is basically the entire value proposition of my event wrap company is find a topic or an event or something that creates Collective attention of a group and have that be the locus and the Catalyst of a wrap because you know the the rap game is entirely predicated on make a thing that is very very widely appreciated and goes viral and is hugely massive and you know making general purpose art is what most rappers rappers are focusing on and I really thought it would be interesting to apply a rapper skill to this uh joyful shared experience and spontaneity Challenge and uh you know thank you for giving me a platform for it and I hope your audience will now imagine in every other possible application of a shared experience that a rapper could add value to because uh if such things occur then my business will succeed and lots of rappers will get lots of gigs rapping in weird places like this and I think that would add to the joyful variety of the world and the genre amazing so thank you so much Baba always welcome to come back give us updates when that rap guide to active inference drops all right I'm on it I'm on it thank you thanks Daniel peace peace